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Faculdade de Tecnologia
Departamento de Engenharia Mecânica

Gated Data-Space Inversion for Reservoir Prediction

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Previsão de Reservatórios usando Inversão do Espaço de Dados Ponderada

Thiago Tomás de Paula

Dissertação de Mestrado submetida ao Programa de Pós-Graduação em Ciências Mecânicas como requisito parcial para obtenção do grau de Mestre.

Orientador: Prof. Dr. Eugênio Libório Feitosa Fortaleza

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Abstract

Accurate prediction of reservoir performance under uncertainty is essential for decision-making in geological reservoir management. Data-Space Inversion (DSI) ([Sun; Durlowsky, 2017](#)) provides an attractive alternative to traditional history matching by updating ensemble forecasts directly in data space, thereby avoiding repeated subsurface simulations. Recently, [Lima, Emerick, and Ortiz \(2020\)](#) introduced DSI-ESMDA, an extension of DSI that employs the Ensemble Smoother with Multiple Data Assimilation (ES-MDA) to replace the Randomized Maximum Likelihood minimizations of the original formulation with an iterative Kalman-type update, achieving significant computational efficiency while maintaining competitive posterior forecasts. Despite these advantages, DSI-ESMDA, like other ensemble smoothers, remains susceptible to excessive variance loss, which can lead to underestimation of posterior uncertainty. Methods such as covariance localization help to ameliorate this issue, but variance underestimation is still observed in practical applications. One contributing factor to this progressive loss of variance is that small production mismatches, often dominated by measurement noise or sampling effects, may trigger repeated micro-updates that reduce ensemble diversity without improving predictive skill. This work introduces an extension to DSI-ESMDA that modifies the effective influence of each observation through a nonlinear, misfit-dependent weighting of the innovation vector. The proposed formulation can be interpreted as introducing a smoothing influence function into the Kalman-type update, computed from normalized innovations using an error-function-based gating operator. In contrast to covariance localization, which acts on second-order statistics, innovation gating acts directly on the update term, selectively suppressing corrections associated with sufficiently-matched observations while preserving the full impact of significant mismatches. Match quality is thresholded by a gating parameter during innovation normalization. The gating methodology is inspired by adaptive filtering developed to mitigate actuator wear in closed-loop control systems ([Fortaleza et al., 2022](#)) and is conceptually related to smooth shrinkage and noise-gating strategies in signal processing ([Donoho; Johnstone, 1994; Donoho, 1995](#)). Although the present work focuses on DSI, the proposed formulation is general and can be applied to standard ensemble-based model-space inversion methods, including ES-MDA and related iterative smoothers. The approach is evaluated on multiple synthetic reservoirs and one real field application using production rates from producers and injectors. Results show that innovation-gated DSI-ESMDA reduces variance loss during and produces more robust uncertainty envelopes in the prediction phase without degrading the data match. Furthermore, for moderate gating parameters, posterior forecasts remain close to those obtained with standard DSI-ESMDA, indicating that the proposed method acts as a stabilizing extension rather than a replacement.

Keywords: ; ; .

Resumo

Palavras-chave: ; ; .

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1 Introduction

1.1 Work overview

The economic success of oil and gas fields is increasingly contingent on the quality of reservoir management decisions, which are made under uncertainty and based on limited data. Kalman-based ensemble methods are widely used to quantify uncertainty in reservoir forecasts by adjusting reservoir models to match production data, but require vast computational resources to perform repeated reservoir simulations. In an effort to reduce the computational burden, Data-Space Inversion (DSI) ([Sun; Durlofsky, 2017](#)) was introduced as an alternative to traditional history matching by updating ensemble forecasts directly in data space, thereby avoiding repeated subsurface simulations.

[Lima, Emerick, and Ortiz \(2020\)](#) introduced DSI-ESMDA, an extension of DSI that employs the Ensemble Smoother with Multiple Data Assimilation (ES-MDA) to replace the Randomized Maximum Likelihood minimizations of the original formulation with an iterative Kalman-type update, achieving significant computational efficiency while maintaining competitive posterior forecasts. However, DSI-ESMDA, like related ensemble smoothers, remains susceptible to excessive variance loss, which can lead to underestimation of posterior uncertainty.

This work introduces an extension to DSI-ESMDA that modifies the effective influence of each observation through a nonlinear, misfit-dependent weighting of the innovation vector. The proposed formulation can be interpreted as introducing a smoothing influence function into the Kalman-type update, computed from normalized innovations using an error-function-based gating operator. Findings on both synthetic and real reservoirs show that Gated DSI-ESMDA reduces variance loss during and produces more robust uncertainty envelopes in the prediction phase without degrading the data match.

Gated DSI-ESMDA acts as a stabilizing extension rather than a replacement, and can be applied to standard ensemble-based model-space inversion methods, including ES-MDA and related iterative smoothers. The power of this contribution is that it provides a simple, yet effective means of improving the robustness of ensemble-based inversion methods without requiring additional simulations or complex modifications to existing workflows.

1.2 Background context

The economic viability of oil and gas operations depends critically on sound reservoir management, where decisions must be made under significant uncertainty and with in-

herently limited subsurface data. Accurate characterization of this uncertainty is essential: overconfident forecasts can lead to poor investment decisions, while excessively uncertain ones impede operational planning. History matching — the process of calibrating reservoir models to observed production data — has long been central to reducing forecast uncertainty, and ensemble-based methods have become the dominant paradigm for performing this calibration at scale.

Kalman-based ensemble methods, such as the Ensemble Smoother with Multiple Data Assimilation (ES-MDA), address this problem by maintaining an ensemble of plausible reservoir models and updating them iteratively as production data are assimilated. By propagating uncertainty through an ensemble rather than relying on a single deterministic model, these methods provide a principled framework for quantifying forecast uncertainty. However, they carry a substantial computational cost: each update cycle requires running full reservoir simulations for every ensemble member, making large-scale uncertainty quantification expensive and, in practice, often infeasible.

Data-Space Inversion (DSI), introduced by Sun and Durlofsky (2017), was proposed as a computationally efficient alternative to traditional model-space history matching. Rather than updating subsurface model parameters and re-running simulations, DSI operates directly on the ensemble of simulated production forecasts — working entirely in data space. This eliminates the need for repeated reservoir simulations during the assimilation process, dramatically reducing the computational burden while still yielding calibrated posterior forecast distributions.

Building on this foundation, Lima, Emerick, and Ortiz (2020) developed DSI-ESMDA, which replaces the Randomized Maximum Likelihood minimizations of the original DSI formulation with an iterative ES-MDA-style Kalman update. This modification brings further gains in computational efficiency and produces competitive posterior forecasts, establishing DSI-ESMDA as a practical and attractive framework for reservoir uncertainty quantification.

Despite these advances, DSI-ESMDA inherits a well-known vulnerability of ensemble smoother methods: excessive variance loss during data assimilation. When observations strongly constrain the ensemble, the spread of forecasts can collapse beyond what is physically justified, leading to an underestimation of posterior uncertainty. This phenomenon is particularly problematic in reservoir forecasting, where robust uncertainty envelopes are essential for risk-aware decision-making. The issue is compounded in data-space methods, where the ensemble is not replenished through new simulations and the initial ensemble spread is the only source of diversity throughout the assimilation process.

Mitigating variance collapse without sacrificing data match quality or introducing additional computational overhead remains an open challenge. Existing remedies — such as inflation schemes and localization — typically require careful tuning and may not transfer cleanly to the data-space setting. This motivates the development of a principled, lightweight

mechanism that can stabilize the Kalman update in DSI-ESMDA, preserving ensemble diversity and producing more reliable uncertainty quantification across a range of reservoir scenarios.

1.3 Research problem

Literature on DSI and DSI-ESMDA take different approaches to mitigating variance loss, but they are either computationally expensive or require careful tuning of hyperparameters. Ideally, a solution would be simple to implement, require minimal tuning, and be computationally efficient. This work proposes introducing a straightforward gating mechanism on DSI-ESMDA's innovation vector, which selectively suppresses the influence of well-matched observations while preserving the impact of significant mismatches. Gated DSI-ESMDA thus modulates the posterior ensemble variance with a single parameter, the intensity of the gating function.

1.4 Research questions

This work addresses the following research questions:

1. Can a gating mechanism be introduced into DSI-ESMDA to selectively modulate the influence of observations based on their match quality?
2. Does the proposed innovation gating reduce variance loss during assimilation and produce more robust uncertainty envelopes in the prediction phase without degrading the data match?
3. How does the performance of Gated DSI-ESMDA compare to standard DSI-ESMDA across synthetic and real reservoir case studies?

1.5 Significance

This work introduces an extension to DSI-ESMDA that modifies the effective influence of each observation through a nonlinear, misfit-dependent weighting of the innovation vector. The proposed formulation can be interpreted as introducing a smoothing influence function into the Kalman-type update, computed from normalized innovations using an error-function-based gating operator. In contrast to covariance localization, which acts on second-order statistics, innovation gating acts directly on the update term, selectively suppressing corrections associated with sufficiently-matched observations while preserving the full impact of significant mismatches. Match quality is thresholded by a gating parameter during innovation normalization. The gating methodology is inspired by adaptive filtering developed to mitigate actuator wear in closed-loop control systems ([Fortaleza *et al.*, 2022](#)) and

is conceptually related to smooth shrinkage and noise-gating strategies in signal processing (Donoho; Johnstone, 1994; Donoho, 1995). Although the present work focuses on DSI, the proposed formulation is general and can be applied to standard ensemble-based model-space inversion methods, including ES-MDA and related iterative smoothers. The proposed method inherits the computational efficiency of DSI-ESMDA while providing a robust, easily tunable mechanism for modulating posterior distribution variance, setting it as a practical and effective tool for uncertainty quantification in reservoir forecasting.

1.6 Methodology overview

The approach is evaluated against the original DSI-ESMDA on two synthetic reservoirs and one real field application using production rates from producers and injectors. The synthetic reservoirs are the Egg and Olympus ensembles, which are widely used in the literature to benchmark history matching and uncertainty quantification methods, while the real field application is a mature oil field in the Brazilian pre-salt. Posterior and prior ensembles are compared across several metrics, including observation coverage and mean squared error, to assess the impact of innovation gating on variance loss and uncertainty quantification. Results show that innovation-gated DSI-ESMDA reduces variance loss during and produces more robust uncertainty envelopes in the prediction phase without degrading the data match. Furthermore, for moderate gating parameters, posterior forecasts remain close to those obtained with standard DSI-ESMDA, indicating that the proposed method acts as a stabilizing extension rather than a replacement.

1.7 Dissertation outline

Chapter 2 presents a review of the literature on reservoir uncertainty quantification, focusing on ensemble-based methods and data-space inversion. Chapter 3 introduces the theoretical framework underpinning the proposed methodology, from the relevant elements of reservoir engineering as a whole to the mathematical formulation of DSI-ESMDA and the innovation gating mechanism. Chapter 4 details the methods used to evaluate the proposed approach, including descriptions of the synthetic and real reservoir case studies, data collection, and analysis techniques. Chapter 5 presents the results of applying innovation-gated DSI-ESMDA to the case studies, including quantitative and qualitative assessments of performance relative to standard DSI-ESMDA. Chapter 6 discusses the implications of the findings and their relationship to the existing literature. Chapter 7 summarizes the main contributions and provides final remarks on the research and future work.

2 Literature Review

2.1 Chapter outline

Aims and structure

The DSI method introduced by Sun and Durlofsky [419] employs PCA to reparameterize the predicted data from the prior ensemble into a lower-dimensional space and uses RML to sample from the posterior distribution of predicted data. The authors applied a data transformation prior to PCA to enhance linearity. Sun et al. [421] further extended the DSI method by introducing a more generalized data transformation procedure, and Sun and Durlofsky [420] applied the method to a geological carbon storage problem. Earlier, Scheidt et al. [381] proposed a similar concept in a method called prediction-focused analysis (PFA), which involves projecting prior predictions into a low-dimensional space and using kernel smoothing to estimate the joint distribution of historical and forecasted data. However, PFA seems to be limited to problems with a small number of data points, as it requires projection into very few dimensions (two or three in the examples presented). Satija and Caers [379] improved the PFA method by incorporating canonical functional component analysis to enhance linearity in the projected data, and Satija and Caers [380] applied this approach to a reservoir problem, finding that it provided uncertainty estimates for production forecasts in reasonable agreement with rejection sampling. Lima et al. [263] combined ES-MDA and DSI, resulting in a more robust and efficient DSI implementation. This method has been further enhanced through the use of various deep learning techniques, leading to successful applications in naturally fractured reservoirs [218] and geological carbon storage [469, 217] (Emerick, 2025).

2.2 Chapter summary

Key takeaways

3 Theory framework

3.1 Chapter outline

Aims and structure

3.2 Elements of reservoir engineering

3.2.1 Reservoir formation

3.2.2 Reservoir simulation

3.3 Kalman Filtering

3.4 Data-Space Inversion

Let $\mathbf{d}$ denote the true production data vector of a reference reservoir across time, and let $\mathbf{d}_j$ denote the production data vector produced by simulating a realization of the reference reservoir using model parameters $\mathbf{m}_j$. For an ensemble of N_e realizations, we have N_e different such vectors, each of which dependent on its own parameters vector. All data vectors are assumed whole, i.e., without missing entries, as well as synchronized.

Of the original reference data vector, only part of it is used in the history matching process. Denote this vector by $\mathbf{d}_h$. Following the same pattern, $\mathbf{d}_{h,j}$ denotes the history match period data output by member j of the ensemble. Assume $\mathbf{d}_h$ is corrupted by additive measurement noise $\mathbf{e}_d \sim N(\mathbf{0}, \mathbf{C}_e)$, where $\mathbf{C}_e$ is the noise covariance matrix. The resulting vector is the observed data during the history period, denoted $\mathbf{d}_{\text{obs}}$. DSI-ESMDA updates the ensemble outputs by iterating a Kalman-like data assimilation, shown below (Lima; Emerick; Ortiz, 2020):

$$\mathbf{d}_j^{k+1} = \mathbf{d}_j^k + (\mathbf{R} \circ \mathbf{K}^k)(\mathbf{d}_{\text{obs}} + \sqrt{\alpha_k} \mathbf{e}_j^k - \mathbf{d}_{h,j}^k), \quad (3.1)$$

with $1 \leq k \leq N_a$, N_a the number of assimilations DSI-ESMDA must do. Thus, $\mathbf{d}_j^k$ for $k = 1$ is the ensemble prior $\mathbf{d}_j$, and $\mathbf{d}_j^{N_a+1}$, the posterior. Vector $\mathbf{e}_j^k \sim N(\mathbf{0}, \mathbf{C}_e)$ is a random disturbance, and α_k coefficients are positive constants that satisfy $\sum_k 1/\alpha_k = 1$.

Matrices $\mathbf{R}$ and $\mathbf{K}$ are the localization and the modified Kalman gain, respectively. $\mathbf{K}$ is defined as

$$\mathbf{K}^k = \Delta \mathbf{D}^k (\Delta \mathbf{D}_h^k)^\top \left(\Delta \mathbf{D}_h^k (\Delta \mathbf{D}_h^k)^\top + \alpha_k \mathbf{C}_e \right)^{-1}, \quad (3.2)$$

with

$$\Delta \mathbf{D} = \frac{1}{\sqrt{N_e - 1}} \begin{bmatrix} \mathbf{d}_1 - \bar{\mathbf{d}} & \cdots & \mathbf{d}_{N_e} - \bar{\mathbf{d}} \end{bmatrix}, \quad (3.3)$$

$$\bar{\mathbf{d}} = \frac{1}{N_e} \sum_{j=1}^{N_e} \mathbf{d}_j, \quad (3.4)$$

a matrix that quantifies the spread of the ensemble. $\Delta \mathbf{D}_h$ is defined analogously by constraining the above equations to the history matching period:

$$\Delta \mathbf{D}_h = \frac{1}{\sqrt{N_e - 1}} \begin{bmatrix} \mathbf{d}_{h,1} - \bar{\mathbf{d}}_h & \cdots & \mathbf{d}_{h,N_e} - \bar{\mathbf{d}}_h \end{bmatrix}, \quad (3.5)$$

$$\bar{\mathbf{d}}_h = \frac{1}{N_e} \sum_{j=1}^{N_e} \mathbf{d}_{h,j}. \quad (3.6)$$

$\mathbf{R}$ adjusts the Kalman gain, preserving correlated measures while diminishing the impact of spuriously correlated measures. As such, its entries lie in $[0, 1]$, and pointwisely multiply $\mathbf{K}$. Intuitively, a correlation is strong if measures of a same quantity were taken closer in space and/or in time, and weaker otherwise. Denoting this “distance” by δ_{uv} , where u is an index from $\mathbf{d}_j$ and v an index from $\mathbf{d}_h$, $\mathbf{R}$ is given by

$$R_{uv} = r(\delta_{uv}), \quad (3.7)$$

$$r(\delta) = \begin{cases} -\frac{1}{4}\delta^5 + \frac{1}{2}\delta^4 + \frac{5}{8}\delta^3 - \frac{5}{3}\delta^2 + 1, & 0 \leq \delta \leq 1 \\ \frac{1}{12}\delta^5 - \frac{1}{2}\delta^4 + \frac{5}{8}\delta^3 + \frac{5}{3}\delta^2 - 5\delta + 4 - \frac{2}{3}\delta^{-1}, & 1 < \delta \leq 2 \\ 0, & \delta > 2 \end{cases}, \quad (3.8)$$

$$\delta_{uv} = \sqrt{\left(\frac{\Delta x'_{uv}}{L_{x'}}\right)^2 + \left(\frac{\Delta y'_{uv}}{L_{y'}}\right)^2 + \left(\frac{\Delta t_{uv}}{T_c}\right)^2}, \quad (3.9)$$

$$\begin{bmatrix} \Delta x'_{uv} \\ \Delta y'_{uv} \end{bmatrix} = \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix} \begin{bmatrix} \Delta x_{uv} \\ \Delta y_{uv} \end{bmatrix} \quad (3.10)$$

Δx_{uv} and Δy_{uv} are the x and y differences, respectively, between the points in space from where measure u and v were taken. In this context, measurements always come from a well, and so these differences are always distances between wells. Δt_{uv} is the difference between the time when measurement u was taken, and the time when v was taken. Note that while v is restrained to the history matching period by definition, u is not. The inclusion of Δt_{uv} guarantees that measurements far apart in time will hold little correlation, even if taken from the same well.

A rotation by θ is applied to take into account the anisotropy of reservoir permeability, and are $L_{x'}$, $L_{y'}$, T_c positive constants that regulate the critical points of differences in x , y and t separately. By critical, it is understood the inflection point $\delta = 1$ in $r(\delta)$. The greater

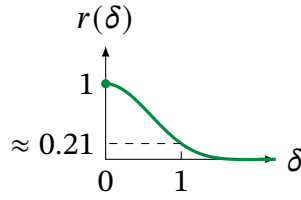


Figure 3.1 – Compact correlation function r used to build the localization matrix $\mathbf{R}$.

Figure 3.2 – Illustration of the localization applied in several different measurements. *Left panel:* Localization between measurements from the same well but from different times. *Center panel:* Localization between measurements from the different wells but at the same times. *Right panel:* Localization between measurements from the different wells at different times.

the critical value, the more lax the localization, allowing more weakly correlated changes to hold more impact.

Localization greatly improves the quality of the posterior, but if the prior zeroes in on an average that is too different from the reference, it is hard to adjust it in order to inject some variance on the collapsed curves. This is partly the reason behind the random disturbance in: injecting variance on the ensemble fit.

Even when the localization $\mathbf{R}$ is well tuned (i.e., θ , $L_{x'}$, $L_{y'}$ and T_c generate adequate posteriors), DSI-ESMDA may suffer the inherent history matching problem of making poor predictions despite good fit to past data. The ensemble posterior narrows around a curve that is not representative of the reservoir.

The inherent problem of history matching may be impossible to solve, but it may be mitigated by allowing the posterior to spread more. We propose doing so by adaptively adjusting the importance of the history mismatch when running the update iterations:

$$\mathbf{d}_j^{k+1} = \mathbf{d}_j^k + (\mathbf{R} \circ \mathbf{K}^k)[\alpha_j^k \circ (\mathbf{d}_{\text{obs}} - \mathbf{d}_{h,j}^k)], \quad (3.11)$$

$$\alpha_{j,v}^k = \text{erf} \left(\frac{1}{\lambda \sqrt{2\sigma_v^2}} |(\mathbf{d}_{\text{obs}} - \mathbf{d}_{h,j}^k)_v| \right) \quad (3.12)$$

The equation above is based on the work of (Fortaleza *et al.*, 2022), and a detailed derivation is provided in [appendix A](#). The smaller the mismatch of a measurement relative to the noise at that measure, the smaller the update. Constant $\lambda \geq 1$ inflates the deviation, further softening the Kalman-like update. At the extreme ($\lambda \rightarrow \infty$ or $\mathbf{d}_{\text{obs}} = \mathbf{d}_{h,j}^k$), $\alpha_j^k = \mathbf{0}$ and the DSI update is fully undone.

3.5 Chapter summary

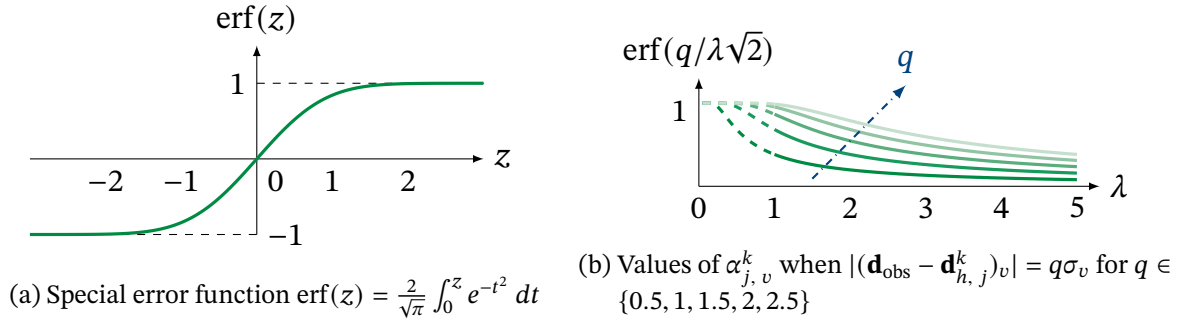


Figure 3.3 – Important functions for understanding the proposed Data-Space Inversion algorithm.

Figure 3.4 – Illustration of the effect of λ on the posterior ensemble for fixed $\mathbf{R}$ and N_a . Yellow lines are the values of $\alpha_j^{N_a+1}$ across time. *Left panel*: DSI-ESMDA prior and posterior ensembles. *Center panel*: DSI-ESMDA with $\lambda = 1$ filtering. *Right panel*: DSI-ESMDA with $\lambda = 5$ filtering. Posterior gains variance as λ grows.

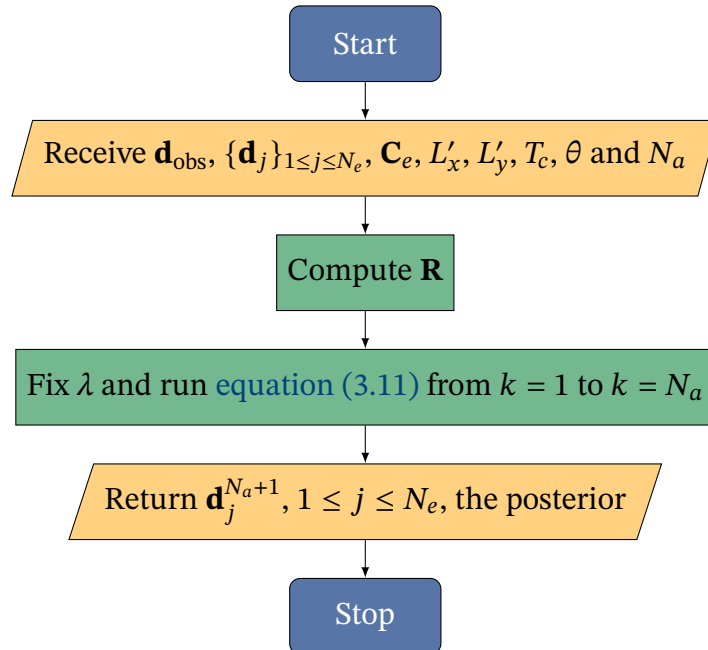


Figure 3.5 – Outline of the proposed Data-Space Inversion algorithm.

Key takeaways

4 Methods

4.1 Chapter outline

Aims and structure

4.2 Reservoir case studies

4.2.1 Egg ensemble

4.2.2 Olympus ensemble

4.2.3 Real field

4.3 Chapter summary

Key takeaways

5 Results

5.1 Chapter outline

Aims and structure

5.2 Egg ensemble

5.3 Olympus ensemble

5.4 Real field

5.5 Chapter summary

Key takeaways

6 Discussion

6.1 Chapter outline

Aims and structure

6.2 Summary of Key Findings

6.3 Interpretation of Findings

6.4 Evaluation of Existing Theories and Models

6.5 Limitations and Future Research

6.6 Chapter summary

Key takeaways

7 Conclusion

7.1 Summary of Findings

7.2 Contribution to Knowledge

7.3 Practical Implications

7.4 Limitations

7.5 Future Research

7.6 Closing Statement

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Appendices

Appendix A – Deriving the Innovation Gating Parameter

This chapter provides a self-contained derivation of the alpha parameter used in the innovation gating mechanism, as introduced in [equation \(3.12\)](#) of [chapter 3](#). The derivation follows closely the work of [Fortaleza *et al.* \(2022\)](#), from where the gating mechanism was inspired.

Let $Y \in \mathbb{R}$ be a random variable representing the output of a given process in time, and Y_I the observation of Y . It is assumed that the observation is affected by a Gaussian noise $N \sim \mathcal{N}(0, \sigma_n^2)$, such that

$$Y_I = Y + N. \quad (\text{A.1})$$

Let y_I be a realization of Y_I . Denoting by y_M the prediction of y from a given model, y_O is defined to be the adjusted prediction after the observation y_I is taken into account. For the sake of simplicity, y_O is given by the following exponential moving average (EMA) relationship:

$$\begin{cases} y_O[0] = y_I[0], & k = 0, \\ y_O[k] = \alpha[k]y_I[k] + (1 - \alpha[k])y_M[k - 1], & k > 0 \end{cases} \quad (\text{A.2})$$

where $\alpha[k] \in [0, 1]$ is the smoothing factor of the EMA at time step index k .

Parameter $\alpha[k]$ represents the statistical confidence that the filter output reflects the same direction of the process output at sample k . More precisely, $\alpha[k]$ is given by the following expression:

$$\begin{aligned} \alpha[k] = & P(\text{sign}(Y_M[k - 1] - Y_I[k]) = \text{sign}(\mu_{Y_M[k-1]} - \mu_{Y_I[k]})) \\ & - P(\text{sign}(Y_M[k - 1] - Y_I[k]) \neq \text{sign}(\mu_{Y_M[k-1]} - \mu_{Y_I[k]})), \end{aligned} \quad (\text{A.3})$$

where $Y_M[k - 1]$ is an observation of Y_M with mean $\mu_{Y_M[k-1]}$ and Y_I an observation of Y_I with mean $\mu_{Y_I[k]}$. Here $\text{sign}(x)$ represents the sign function, which is defined as:

$$\text{sign}(x) = \begin{cases} 1, & x > 0, \\ 0, & x = 0, \\ -1, & x < 0. \end{cases} \quad (\text{A.4})$$

[Equation \(A.3\)](#) can be rewritten as follows:

$$\alpha[k] = 2P(\text{sign}(Y_M[k - 1] - Y_I[k]) = \text{sign}(\mu_{Y_M[k-1]} - \mu_{Y_I[k]})) - 1. \quad (\text{A.5})$$

Defining a random variable Y_{MI} whose observation at time k have mean and variance given by

$$\mu_{Y_{MI}[k]} = |\mu_{Y_M[k-1]} - \mu_{Y_I[k]}|, \quad (\text{A.6})$$

$$\sigma_{Y_{MI}[k]}^2 = \sigma_{Y_M[k-1]}^2 + \sigma_{Y_I[k]}^2 - 2\text{Cov}(Y_M[k-1], Y_I[k]), \quad (\text{A.7})$$

where $\text{Cov}(Y_M[k-1], Y_I[k])$ is the covariance between $Y_M[k-1]$ and $Y_I[k]$, [equation \(A.5\)](#) can be expressed as follows:

$$\alpha[k] = 2P(Y_{MI}[k] > 0) - 1. \quad (\text{A.8})$$

In order to find a practical implementation of $\alpha[k]$, [equation \(A.8\)](#) is analyzed under the following technical assumptions.

Assumption A.1. The random variable $Y_{MI}[k]$ can be approximated by a normal distribution with mean $\mu_{Y_{MI}[k]}$ and variance $\sigma_{Y_{MI}[k]}^2$.

Assumption A.2. Both high frequency and random behavior of the measurement noise implies that the observations $Y_M[k-1]$ and $Y_I[k]$ can be considered approximately statistically independent of each other. Mathematically, $\text{Cov}(Y_M[k-1], Y_I[k]) \approx 0$.

Assumption A.3. The variance of the model's prediction is much smaller than the variance of the measured observation. Mathematically, $\sigma_{Y_M[k-1]}^2 \ll \sigma_{Y_I[k]}^2$.

Assumption A.4. By considering an additive zero-mean measurement noise, the samples $y_M[k-1]$ and $y_I[k]$ can be used to estimate $\mu_{Y_M[k-1]}$ and $\mu_{Y_I[k]}$, respectively.

From [assumption A.1](#), [equation \(A.8\)](#) can be rewritten as:

$$\begin{aligned} \alpha[k] &= 2P(\mathcal{N}(\mu_{Y_{MI}[k]}, \sigma_{Y_{MI}[k]}^2) > 0) - 1 \\ &= 2P(\mathcal{N}(0, \sigma_{Y_{MI}[k]}^2) \leq \mu_{Y_{MI}[k]}) - 1, \end{aligned} \quad (\text{A.9})$$

which can be expressed in terms of the Gauss error function:

$$\begin{aligned} \alpha[k] &= \text{erf}\left(\frac{\mu_{Y_{MI}[k]}}{\sqrt{2\sigma_{Y_{MI}[k]}^2}}\right) \\ &= \text{erf}\left(\frac{|\mu_{Y_M[k-1]} - \mu_{Y_I[k]}|}{\sqrt{2[\sigma_{Y_M[k-1]}^2 + \sigma_{Y_I[k]}^2 - 2\text{Cov}(Y_M[k-1], Y_I[k])]}}\right). \end{aligned} \quad (\text{A.10})$$

Applying [assumption A.2](#), [assumption A.3](#) and [assumption A.4](#), [equation \(A.10\)](#) becomes

$$\alpha[k] = \operatorname{erf} \left(\frac{|y_M[k-1] - y_I[k]|}{\sqrt{2\sigma_{Y_I[k]}^2}} \right) = \operatorname{erf} \left(\frac{|y_M[k-1] - y_I[k]|}{\sqrt{2\sigma_n^2[k]}} \right). \quad (\text{A.11})$$

Since knowledge of the measurement noise variance $\sigma_n^2[k]$ is not always available in practice, a fixed $\lambda \geq 1$ is introduced to scale the measurement noise standard deviation. Thus, the final expression for $\alpha[k]$ is given by

$$\alpha[k] = \operatorname{erf} \left(\frac{|y_M[k-1] - y_I[k]|}{\lambda \sqrt{2\sigma_n^2[k]}} \right) \quad (\text{A.12})$$

which is equivalent to [equation \(3.12\)](#) in [chapter 3](#).



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